Teaching on Borrowed Time: How the Teacher Loan Forgiveness Program Delays, But Does Not Prevent, Teacher Attrition

Seth Walker*

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Abstract

Service-based student loan programs are widely used to encourage teachers to work in high-need schools and subject areas, yet relatively little is known about how teachers' careers evolve beyond required service periods. Using administrative data from Texas and differences in teacher loan eligibility and initial school eligibility, this paper studies employment and earnings outcomes surrounding the five-year service requirement of the Teacher Loan Forgiveness Program (TLFP). I find no evidence of higher retention during the service period itself. Instead, exits from Texas public schools (TXPS) rise after the five-year threshold, with responses concentrated among teachers with higher student loan balances. Among teachers who remain in TXPS, school-level retention also declines, indicating increased mobility within the system, though without systematic shifts in school characteristics, earnings, or labor supply. Taken together, the results indicate that the TLFP primarily alters the timing of employment decisions within TXPS rather than generating sustained improvements in retention, highlighting the limitations of one-time, service-based incentives as a long-term teacher workforce strategy.

*Department of Economics, Michigan State University. Email: walke893@msu.edu

1 Introduction

Teacher shortages are a persistent challenge in U.S. education, particularly in high-need schools and fields such as special education, mathematics, and science (Garcia and Weiss, 2020). Recent estimates show a minimum of 36,000 vacant teaching positions and 163,000 roles are filled by underqualified educators, with low-income schools and high-need subject areas bearing the brunt of the crisis (Carver-Thomas et al., 2021; Nguyen et al., 2022). Low salaries combined with the rising costs of higher education have led to declining enrollment in teacher preparation programs and high turnover rates that have exacerbated teacher shortages, further contributing to staffing instability (Podolsky and Kini, 2016; Ingersol et al., 2018; Ingersoll et al., 2019; Sargrad et al., 2019; Loeb and Myung, 2020; Wilson and Kelley, 2022).

The consequences of teacher turnover extend beyond staffing challenges. High attrition disrupts student learning, with research linking frequent teacher changes to lower academic achievement (Ronfeldt et al., 2013). Moreover, teacher turnover imposes significant financial costs on school districts—replacing a single teacher costs an average of $25,000, reflecting recruitment, hiring, and training expenses (Sutcher et al., 2019; Learning Policy Institute, 2024). In high-turnover districts, the financial burden diverts resources from student-focused initiatives, further widening educational inequities.

To address these challenges, policymakers have implemented numerous initiatives aimed at reducing the financial costs of entering the teaching profession and incentivizing teachers to remain in the profession. A central motivation for these efforts is the financial strain faced by new teachers: more than half borrow to finance their education, with average balances exceeding $50,000 (Herschcopf et al., 2021; Garcia et al., 2023). With relatively low salaries, these debt burdens heighten the appeal of financial incentives tied to service, making loan forgiveness a particularly salient policy lever for recruitment and retention. In line with these concerns, as of July 2023, 18 states offer teacher-specific loan forgiveness programs (Isola, 2023). Many states also provide a range of other incentives, including annual bonuses, retention incentives, mortgage assistance, and tax credits.

In addition to state-level initiatives, large federal student loan forgiveness programs are also available to teachers. Programs such as the Teacher Loan Forgiveness Program (TLFP) and Public Service Loan Forgiveness (PSLF) provide student loan forgiveness to eligible public-sector workers, including teachers, conditional on meeting service requirements. Together, these state and federal initiatives form an important part of the policy landscape facing the teacher workforce, underscoring the importance of understanding their effects over teachers' careers.

This paper studies how teachers' employment decisions change following the five-year service requirement of the TLFP. The TLFP and similar programs are intended to encourage teachers to remain in high-need schools and subject areas, yet little is known about whether they affect teachers' employment decisions beyond the required service period. By focusing on outcomes after the five-year service window, this paper provides new evidence on the extent to which the TLFP promotes sustained retention in public schools.

Using administrative data from Texas, I show that teachers eligible for the TLFP are more likely to leave Texas public schools (TXPS) after the five-year service period, with effects concentrated among teachers with large student loan balances. For teachers who remain in TXPS, school-level retention also declines at this point. However, I find no evidence that this increased mobility corresponds to systematic shifts in school characteristics or to changes in earnings or labor supply. Together, these findings indicate that the TLFP primarily shifts the timing of employment decisions rather than producing durable improvements in retention within TXPS.

The remainder of the paper proceeds as follows. Section 2 provides background on the Teacher Loan Forgiveness Program, while Section 3 situates the paper within the literature on student debt and financial incentives for teachers. Section 4 describes the data used in the analysis. Section 5 presents the empirical framework and discusses the results. Section 6 concludes with a discussion of the findings, their implications for teacher retention policies, and directions for future research.

2 Teacher Loan Forgiveness Program

The Teacher Loan Forgiveness Program (TLFP), coordinated by the U.S. Department of Education, aims to address teacher shortages by incentivizing educators to work in low-income schools and high-need subjects. The program provides up to $17,500 in federal student loan forgiveness to teachers who meet professional qualification requirements and complete a period of service. Teachers in secondary mathematics, secondary science, and special education (SPED) are eligible for the maximum $17,500 in forgiveness, reflecting higher demand in these fields, while all other eligible teachers qualify for $5,000. The program applies to teachers with federal subsidized or unsubsidized loans issued after October 1st, 1998.

To qualify, teachers must hold a bachelor's degree, obtain full state certification, and demonstrate subject-area competence.[1] For elementary school teachers, this requires proficiency in reading, writing, mathematics, and the core elementary curriculum, while secondary school teachers must demonstrate expertise in their assigned subject through testing, degree completion, or advanced certification.

The service requirement consists of five consecutive years of full-time teaching in low-income schools or educational service agencies listed on the Teacher Cancellation Low Income (TCLI) Directory. States submit qualifying schools annually to the Department of Education, typically using a cutoff of at least 30% of student enrollment from low-income or economically disadvantaged families.[2] The service period must be completed consecutively, though years served continue to count if a school later loses eligibility while a teacher remains employed at that school. Upon completing the service period and making at least 60 qualifying monthly loan payments, teachers must submit

[1] Certification and licensure requirements cannot be met by emergency, temporary, or provisional measures.

[2] Schools operated by the Bureau of Indian Education or that are in districts receiving Title I funds also qualify. Private schools do not qualify unless they are 501(c)(3) nonprofits, participate in Title I, and meet the state's low-income threshold. In 2017–18, 24.3% of private schools participated in Title I, with participating schools averaging 42% of students eligible for free or reduced-price lunches, leaving relatively few private schools eligible for TLFP forgiveness (Taie and Goldring, 2019).

a certified Teacher Loan Forgiveness Application to each loan servicer to receive loan forgiveness.

Although the TLFP is intended to improve teacher retention, its reach has been limited in practice. Since its inception in 1998, the program has forgiven loans for more than 200,000 teachers, totaling over $3 billion in loan forgiveness. Prior work documents substantial administrative frictions in the program's implementation, including complex application procedures and servicing errors that delay or deny forgiveness for otherwise eligible teachers (Nowicki, 2015).

The TLFP also operates alongside other student loan forgiveness programs, most notably Public Service Loan Forgiveness (PSLF). Under PSLF, borrowers may receive forgiveness of remaining federal loan balances after ten years of qualifying public service and 120 qualifying payments. Because service years cannot count simultaneously toward both programs, some teachers who are eligible for the TLFP may pursue PSLF, particularly those with larger loan balances. Nevertheless, PSLF accounts for only 12.2% of all federal student loan forgiveness, while the teacher-specific TLFP accounts for 22.1% (Hanson, 2024). Thus, despite overlap with PSLF, the TLFP remains the primary teacher-specific loan forgiveness program and a central component of the policy environment facing teachers.

3 Literature Review and Contributions

3.1 Literature Review

A large literature on teacher labor markets emphasizes compensation as a central driver of recruitment and retention, particularly in high-need schools and subject areas. Low salaries are strongly associated with early-career attrition (Ashiedu and Scott-Ladd, 2012; Dee and Goldhaber, 2017), especially in low-income schools (Ingersoll, 2004; Ingersoll and May, 2012; Podolsky and Kini, 2016; Garcia and Weiss, 2020). Although permanent salary increases can improve retention (Feng, 2009), their costs have led policymakers to increasingly rely on alternative compensation mechanisms.

A growing body of work examines the effects of short-term incentive programs on teacher retention. Clotfelter et al. (2008) find that an annual $1,800 bonus reduced turnover among certified math, science, and SPED secondary teachers in targeted schools by 17%. Similarly, Cowan and Goldhaber (2018) report a 31-41% reduction in turnover from a $5,000 annual bonus for teachers in high-poverty schools in Washington State. Glazerman et al. (2013) show that selective transfer incentives of $20,000 paid over two years increased retention during the payout window, but that these effects dissipated once payments ended.

These studies show that temporary compensation incentives can reduce teacher turnover, though evidence on whether these effects persist beyond the incentive period is limited. Most existing work, however, focuses on compensation delivered through wages or bonuses, with comparatively less attention paid to compensation provided through student loan forgiveness or repayment assistance. Related research shows that reductions in debt increase mobility (Di Maggio et al., 2020), while higher debt burdens push individuals toward higher-paying occupations (Rothstein and Rouse, 2011; Luo and Mongey, 2019). Together, these findings suggest that debt forgiveness can meaningfully influence employment decisions, making it a potentially important yet understudied policy lever.

Survey and administrative evidence indicate that student loan forgiveness is both widespread and behaviorally salient. Recent survey data from the Consumer Financial Protection Bureau show that roughly 22% of federal student loan borrowers have applied for some form of debt relief (Caldwell et al., 2024). Among borrowers who received forgiveness, 11% report that it enabled them to move, 9% to change jobs or start a business, and nearly half indicate that it allowed them to save more, pointing to meaningful effects on mobility, job choice, and financial behavior.

Complementary evidence comes from Dinerstein et al. (2025), who find that student loan forgiveness leads to a short-run decline in earnings of roughly $44 per month (2.3%), consistent with reduced labor supply following forgiveness. The same study documents increased transitions out of public service, in line with forgiveness relaxing job lock created by service-based eligibility requirements. Together, these findings underscore that loan forgiveness can affect not only when individuals change jobs, but also how much they work and where they are employed.

Studies on service-oriented professions show that student loan repayment and forgiveness programs shape job choice and retention decisions. Pathman et al. (2004) find that loan repayment programs for physicians encourage practice in lower socioeconomic areas and increase retention in those settings, while Scheckel et al. (2019) show that positions eligible for loan forgiveness attract greater interest, particularly among physicians with higher debt levels. Survey evidence from social workers in Massachusetts similarly indicates that loan forgiveness is perceived as a factor reducing turnover intentions, highlighting its salience for workforce retention decisions (Fakunmoju and Kersting, 2016). Finally, Field (2009) studies the random assignment of financial aid packages to law students and find that those offered tuition waivers in exchange for committing to public-interest work were 36% more likely to enter public-interest law than students offered loan forgiveness after completing such service.

Turning to educators specifically, survey results from TLFP recipients indicate that loan forgiveness shapes both pre-career and within-career decisions. Among teachers who received forgiveness, 15% report that they moved, 50% report increased saving, and 4% report changing jobs after receiving relief (Caldwell et al., 2024). In addition, 20-25% of applicants report that the program influenced their major, career choice, or employer selection, pointing to effects on both retention and initial entry into the teaching profession.

Additional research from teacher-focused loan repayment and forgiveness programs shows that these incentives shape educators' career decisions within the profession. The North Carolina Teaching Fellows program, which provides forgivable loans for teaching in STEM or special education fields, exhibits high retention during the commitment period but lower retention after service requirements end (Henry et al., 2012). Likewise, survey evidence from Noyce Scholarship recipients indicates that 70% report the program influenced their decision to teach in—and remain at—a high-need school through the required service commitment (Podolsky and Kini, 2016).

Estimated effects on teacher retention from program evaluations are more mixed. Feng and Sass

(2018) find that Florida's Critical Teacher Shortage Program, which offered \$10,000 in loan repayment staggered over four years, reduced middle and high school mathematics and science teacher turnover by 8.9-10.4%. In contrast, Russell (2020) examine school-level retention rates around the 30% threshold for Teacher Loan Forgiveness Program (TLFP) eligibility in Massachusetts, New York, North Carolina, and South Carolina, and find no difference in retention between marginally eligible and ineligible schools.

Most closely related to this study is the analysis of school-level TLFP eligibility using teacher-level data from Michigan conducted by Jacob et al. (2024). By focusing on teacher retention outcomes and restricting the sample to new teachers, the authors study a population particularly relevant for examining loan forgiveness incentives. Using a regression discontinuity design that compares teachers whose first-year schools fall just above or below the 30% TLFP eligibility threshold, they show that teachers do not sort into eligible schools, with no significant discontinuities in teacher demographics or school characteristics at the cutoff. They further estimate no differences in retention during years two through five and interpret these results as evidence that school eligibility does not affect teachers' employment decisions. However, in a contingent valuation survey, the authors find that teachers place a positive value on working in TLFP-eligible schools. Average valuations are approximately \$500, rising to about \$3,000 when eligibility is made explicit, with larger valuations among teachers with federal student loans and those eligible for higher forgiveness amounts.

3.2 Contributions

This paper builds on recent work examining the TLFP by extending the analysis along several important dimensions. Whereas prior work focuses on school-level eligibility for the program, this study incorporates detailed teacher-level information to more precisely capture eligibility criteria. Using linked administrative data from Texas that connect teachers to their student loan records, I identify which teachers hold eligible federal loans. In addition, certification, licensing, and degree records allow me to determine when teachers meet the program's professional qualification requirements. Together, these data enable a more accurate characterization of eligibility and support cleaner estimates of the effects of program eligibility at the teacher level.

A second contribution is the extension of the time horizon over which outcomes are evaluated. Existing studies primarily examine retention during the five-year service requirement window. In contrast, this analysis considers outcomes both during and after the service requirement period, allowing for an assessment of longer-run responses. This distinction is important for understanding whether service-linked incentives generate persistent changes in workforce attachment or primarily delay attrition until incentives expire.

Beyond retention, this study examines a broader set of labor market outcomes. For teachers who remain, I analyze changes in supplemental earnings and engagement in employment outside the classroom, capturing adjustments in work activity beyond their primary teaching role. For teachers who exit, I examine subsequent earnings and sector of employment, shedding light on where teachers work after leaving TXPS. Together, these outcomes provide a more comprehensive view of how program incentives shape teachers' labor supply and career trajectories beyond retention.

Finally, this study examines how responses to the Teacher Loan Forgiveness Program vary across teachers. I explore heterogeneity along several dimensions, with particular emphasis on differences in student loan balance and the amount of loan forgiveness for which teachers are eligible. Documenting this variation helps inform how loan-forgiveness policies may differentially affect teachers facing different financial circumstances.

4 Data

Data for this paper come from the Houston Education Research Center, which integrates detailed longitudinal data from multiple state agencies. Specifically, these include postsecondary records from the Texas Higher Education Coordinating Board (THECB), certification files from the State Board for Educator Certification (SBEC), public school employment records from the Texas Education Agency (TEA), and quarterly earnings reported to the Texas Workforce Commission (TWC). Together, these sources allow me to track teachers' education, certification, employment, and earnings trajectories before and after the TLFP's five-year service requirement.

The THECB files contain complete financial aid histories, including all grants, scholarships, and federal and state student loans received. I use these data to identify teachers with loans eligible for TLFP forgiveness and to measure loan balances. The files also report graduation dates, institutions, and majors, which I use to verify degree completion and subject-area alignment with TLFP eligibility criteria. The SBEC files provide detailed information on teachers' certification, including issuance dates, certification type (standard, emergency, provisional, or temporary), program field, and certifying organization and route.

Employment data from the Texas Education Agency (TEA) document every public school position held by educators in Texas and report the associated school, district, grade level, subject, and full-time equivalency (FTE). From these records, I construct annual indicators for employment in TXPS, as well as school- and district-level retention. The TEA data also allow me to determine whether teachers qualify for $5,000 or $17,500 in TLFP forgiveness based on their subject area and grades taught. Finally, the files report base pay and supplemental earnings, capturing compensation for additional duties such as athletics, clubs, performing arts, or tutoring.

The TWC data provide quarterly wage reports from employers covered by Texas Unemployment Insurance and include the corresponding North American Industry Classification System (NAICS) industry codes. These files allow me to measure outside earnings, employment sector, and transitions out of the public school system. All earnings are adjusted to 2024 dollars using the Bureau of Labor Statistics' Consumer Price Index for All Urban Consumers (Series ID: CUUR0000SA0).

By combining TEA and TWC records, I construct a proxy for employment in private schools using NAICS industry codes and public school employment records. I classify an individual as employed in a private school if they do not appear in the TEA's public school employment records and report earnings in the NAICS industry group (four-digit) or national industry (six-digit) code for "Elementary and Secondary Schools."[3] I also construct two additional outcomes: (1) exit from teaching, defined as no longer working in either a public or private school but remaining employed in Texas, and (2) exit from the Texas workforce, defined as having no reported earnings for an entire year.

For my analysis, I restrict the sample to teachers who earned a bachelor's degree from a Texas public university and whose first year of teaching was full-time and fully certified in a Texas public school between the 2000-01 and 2013-14 academic years. This restriction aligns the analytic sample with the population for whom the TLFP service requirement is well-defined and most salient, as eligibility requires five consecutive years of full-time, fully certified teaching. Teachers who enter under provisional certification or part-time appointments often follow different certification timelines and exhibit higher early-career churn, making it difficult to distinguish program-induced retention from exits driven by credentialing or employment constraints. Excluding these individuals focuses the analysis on teachers who began their careers as fully credentialed educators and can be observed for at least eight years after entry, capturing both the TLFP's five-year service window and subsequent outcomes.

I further restrict the analytic sample to teachers employed at regular instructional campuses. Teachers whose initial placements were in alternative settings, such as juvenile justice facilities, disciplinary alternative programs, residential or hospital schools, virtual-only campuses, and special-purpose academies, are excluded. Although some of these campuses appear on the TCLI Directory, they serve highly specialized student populations and operate under staffing models that differ from those of standard school environments. This restriction provides a more consistent basis for comparing teachers' career trajectories and assessing the TLFP's effects.

Panel A of Table 1 summarizes demographic and entry characteristics for the 87,012 teachers in the resulting analytic sample. Teachers enter the profession at an average age of 26, and the workforce is predominantly female (85%) and White (64%), with Hispanic/Latino teachers comprising 28% and Black and Asian teachers accounting for 5% and 2%, respectively. This composition closely mirrors the statewide teacher workforce during these years (Landa, 2024). Nearly nine in ten teachers enter through traditional certification routes, a higher share than the statewide average. This pattern reflects the requirement that teachers' first year be full-time and fully certified, which excludes many alternative-route entrants who begin teaching under temporary or provisional licenses. 30% hold an advanced degree, and just over two-thirds have loans eligible for forgiveness under the TLFP.

Panel B restricts to the 68% of teachers with loans eligible for TLFP forgiveness. The average eligible loan balance is \$44,718 (2024 dollars), with balances exceeding \$9,531 for 90% of teachers, \$54,331 for one-quarter, and \$80,522 for one-tenth. Most eligible teachers (89%) qualify for \$5,000 in forgiveness, while 11% qualify for \$17,500 based on their subject area and grade assignment. Among those eligible for the higher forgiveness amount, approximately two-thirds are secondary mathematics or science teachers, with the remaining one-third in special education. Finally, 83%

[3] See NAICS (2024) for a detailed breakdown of NAICS codes.

Panel A: Teacher Descriptives		Panel B: Eligible Loan Teacher Descriptives	
Outcome	**Value**	**Outcome**	**Value**
Starting Age	25.98	TCLI Directory	0.83
White	0.64	$17,500 Forgiveness Eligibility	0.11
Hispanic/Latino	0.28	**Eligible Loan Distribution**	
Black	0.05	10th Percentile	$9,531
Asian	0.02	25th Percentile	$19,629
Female	0.85	Median	$33,936
Traditional Certification Route	0.89	Mean	$44,718
Advanced Degree	0.30	75th Percentile	$54,331
TLFP Eligible Loans	0.68	90th Percentile	$80,522
N	87,012	N	59,495

Notes: Panel A includes all TXPS teachers in the analytic sample and reports demographic descriptive statistics. Panel B restricts to to teachers with TLFP-eligible loans. The TCLI Directory indicator equals one if the teacher's first school appears on the TCLI Directory, reflecting initial school eligibility for the TLFP. Panel B also reports forgiveness amount eligibility and the distribution of eligible loan balances. Loan balances are reported in 2024 dollars, deflated using the Bureau of Labor Statistics' College Tuition and Fees Price Index (Series ID: CUUR0000SEEB01).

Table 1. Teacher and Loan Descriptive Statistics

began their careers in schools listed on the TCLI Directory, reflecting the broad reach of the program across TXPS.

5 Empirical Strategy and Results

5.1 Teacher Sorting at Eligibility Threshold

I begin by examining whether teachers sort into TLFP-eligible schools at career entry, an early margin through which loan forgiveness incentives may operate. Specifically, I test whether observable characteristics differ between teachers with and without TLFP-eligible loans around the 30% economically disadvantaged threshold using Equation 1:[4]

$$\begin{aligned} X_i = \alpha + Above_{is,t=1} + Loans_i + \tilde{R}_{is,t=1} + \beta(Above_{is,t=1} \times Loans_i) + \\ (Above_{is,t=1} \times \tilde{R}_{is,t=1}) + (Loans_i \times \tilde{R}_{is,t=1}) + (Loans_i \times Above_{is,t=1} \times \tilde{R}_{is,t=1}) + \epsilon_{is} \end{aligned} \tag{1}$$

[4]For the 2003–04 to 2013–14 school years, 94.2% of schools above the 30% economically disadvantaged threshold appear on the TCLI Directory, compared to 8.7% of schools below it. Appendix Figure A.1 illustrates the sharp increase in inclusion at the threshold. Small deviations from the cutoff reflect reporting anomalies, rounding in small schools, and eligibility rules for certain non-traditional or Bureau of Indian Education-affiliated schools.

where X_i represents an observable teacher characteristic (e.g., race/ethnicity, gender, age, certification route, student loan balance, or forgiveness eligibility amount); $Above_{is,t=1}$ is an indicator for whether teacher i's initial school is above the 30% economically disadvantaged threshold; $Loans_i$ equals one if teacher i has loans eligible for forgiveness under the TLFP; and $\tilde{R}_{is,t=1}$ denotes the centered percentage of economically disadvantaged students at teacher i's initial school. The coefficient of interest, β, captures differences in observable characteristics at the eligibility threshold between teachers with and without eligible loans. Bandwidths are selected using the procedures of Calonico et al. (2014), yielding an optimal bandwidth of 18.64 percentage points (pp). I also report sensitivity checks using alternative bandwidths of 10 and 30 pp.

Variable	**(1) Optimal BW**	**(2) +/-10**	**(3) +/-30**
Forgiveness Amount: $17,500	0.006	-0.013	0.000
	(0.0184)	(0.0258)	(0.0144)
Proportion with TEACH Grant	-0.001	-0.010	0.002
	(0.0076)	(0.0101)	(0.0063)
Advanced Degree	0.000	0.006	-0.002
	(0.0243)	(0.0334)	(0.0192)
Traditional Route	0.004	-0.002	-0.004
	(0.0175)	(0.0243)	(0.0138)
Age	-0.313	-0.168	-0.270
	(0.3367)	(0.4631)	(0.2690)
White	0.042**	0.088***	0.041**
	(0.0206)	(0.0280)	(0.0167)
Black	-0.013	-0.024**	-0.023***
	(0.091)	(0.0123)	(0.0074)
Hispanic	-0.023	-0.053**	-0.010
	(0.0174)	(0.0235)	(0.0142)
Asian	-0.005	-0.008	-0.005
	(0.0073)	(0.0100)	(0.0058)
Female	0.016	0.011	0.007
	(0.0203)	(0.0283)	(0.0159)
N	24,443	12,792	40,219

Notes: This table contains teacher sorting estimates around the 30% economically disadvantaged threshold that determines TLFP school eligibility. Coefficients represent discontinuities in observable teacher characteristics at the threshold. Columns report results using the optimal bandwidth and bandwidths of ±10 and ±30 pp. Standard errors clustered at the initial school level are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

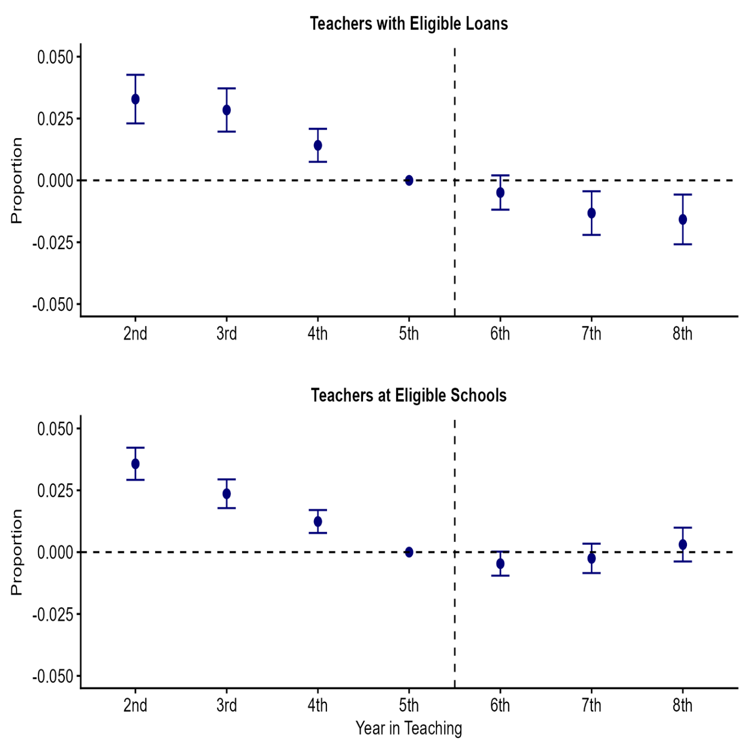
Table 2. Teacher Sorting into Eligible Schools

Table 2 reports estimates of sorting into TLFP-eligible schools around the 30% economically disadvantaged threshold using the optimal bandwidth and the two alternative bandwidths. Across most observable characteristics, estimates are insignificant and close to zero. However, there are patterns of sorting along racial lines primarily for White teachers with eligible loans, who are 4.1-8.8 pp more likely to enter eligible schools at the threshold depending on bandwidth. Estimates for Hispanic and Black teachers are less consistent across specifications and are only statistically significant in the alternative bandwidths. Beyond these patterns, there is little evidence of systematic differences in entry characteristics around the eligibility threshold.

5.2 Difference-in-Difference-in-Differences

I next estimate the impacts of the TLFP using a difference-in-difference-in-differences (DDD) framework that leverages variation in both teacher loan eligibility and initial school eligibility. This design isolates the effect of beginning one's career in a TLFP-eligible school on outcomes after the five-year service window for teachers with eligible loans.

A simpler difference-in-differences (DiD) design is insufficient in this setting, as designs that rely only on loan eligibility or school eligibility generate differential retention trends in the first five years of teaching. The two panels of Figure 1 present event-study estimates from these DiD specifications, showing that outcomes diverge before teachers reach the five-year mark.



Notes: This figure contains event study estimates from the DiD conditioning on teachers having eligible loans (top figure) and teachers starting at an eligible school (bottom figure). Estimates are relative to the final year of the five-year service window (year 5).

Figure 1. DiD Event Studies: Exit TXPS

Specifically, comparing teachers with eligible loans in eligible and ineligible schools (top panel of Figure 1) conflates potential TLFP impacts with baseline differences in retention across school types, as exit rates are higher in eligible schools during the first five years of teaching. Conversely, comparing teachers with and without eligible loans within eligible schools (bottom panel of Figure 1) captures differences associated with loan status, as teachers with eligible loans exhibit higher early-career exit rates independent of school eligibility. Intuitively, the DDD design compares the difference in outcomes for teachers with eligible loans at eligible versus ineligible schools to the corresponding difference for teachers without eligible loans, thereby removing retention patterns that arise from school eligibility or loan status alone.

The DDD specification is outlined below in Equation 2:

$$
\begin{aligned}
Y_{ist} = \alpha + TCLI_{is,t=1} + Loans_i + After5_{it} + (TCLI_{is,t=1} \times Loans_i) + \\
(TCLI_{is,t=1} \times After5_{it}) + (Loans_i \times After5_{it}) + \\
\beta(TCLI_{is,t=1} \times Loans_i \times After5_{it}) + X_{is,t=1} + \delta_{\text{Cohort}(i)} + \lambda_t + \epsilon_{ist}
\end{aligned}
\tag{2}
$$

where $TCLI_{is,t=1}$ is an indicator equal to one if teacher i's initial school s is on the TCLI directory; $Loans_i$ equals one if teacher i has loans eligible for forgiveness under the TLFP; and $After5_{it}$ is an indicator equal to one beginning in teacher i's sixth year of teaching (and in all subsequent years). $X_{is,t=1}$ includes teacher characteristics at entry (race, gender, advanced degrees, age, and certification route) and initial school characteristics (percent economically disadvantaged, charter status, and elementary/secondary designation). Cohort fixed effects ($\delta_{\text{Cohort}(i)}$) capture differences across entering cohorts, while calendar-year fixed effects (λ_t) absorb common shocks to retention and labor market conditions. Standard errors are clustered at the initial school level.

The coefficient of interest, β, captures the post-five-year effect of initial placement in a TLFP-eligible school for teachers with eligible loans. The outcome variable Y_{ist} encompasses multiple dimensions of teacher employment, including retention outcomes (exiting TXPS, moving to a private school, exiting teaching, and school- and district-level retention); school and job characteristics (changes in employing school characteristics and holding multiple positions within or outside teaching); and labor market outcomes (annual earnings and sector of employment).

Importantly, the $After5_{it}$ indicator does not require teachers to remain at the same school or at a TLFP-eligible school for five consecutive years, as conditioning on such continuity would introduce selection bias through post-entry mobility decisions.[5] Additionally, because I cannot observe which teachers apply for or receive loan forgiveness, estimates reflect the effect of potential eligibility associated with initial placement at a TLFP-eligible school after five years.

As with other DiD approaches, the validity of this approach rests on a parallel trends assumption. In this setting, absent the TLFP, the difference in retention between eligible and ineligible schools would have evolved similarly for teachers with and without eligible loans (Olden and Møen, 2022). To assess the plausibility of this assumption, I estimate event studies using the specification in Equation 3.

$$
\begin{aligned}
Y_{ist} = \alpha + TCLI_{is,t=1} + Loans_i + (TCLI_{is,t=1} \times Loans_i) + \\
\sum_{j=-4}^{3} (D_{ij} \times TCLI_{is,t=1}) + \sum_{j=-4}^{3} (D_{ij} \times Loans_i) + \\
\sum_{j=-4}^{3} \kappa_j (D_{ij} \times TCLI_{is,t=1} \times Loans_i) + X_{is,t=1} + \delta_{\text{Cohort}(i)} + \lambda_t + \epsilon_{ist}
\end{aligned}
\tag{3}
$$

[5] Among teachers with eligible loans who start their careers at a TLFP-eligible school and remain in TXPS for at least five years, 88.0% teach at an eligible school for the full five years. In contrast, 84.9% of teachers without eligible loans remain at an eligible school during their first five years in TXPS.

where j indexes years relative to the end of the five-year service window and D_{ij} is an indicator for event time. The coefficients of interest, κ_j, capture the dynamic effects of initial placement in a TLFP-eligible school for teachers with eligible loans, relative to teachers without eligible loans. These coefficients are plotted in event-study graphs to examine dynamic patterns before and after the five-year service window.

5.2.1 Employment

Table 3 reports estimates from Equation 2 on teachers' employment outcomes. Panel A reports effects on overall employment patterns five years after entry. Teachers with eligible loans who began in TLFP-eligible schools are 1.2 pp more likely to exit TXPS, approximately an 11% increase relative to the baseline exit rate. This effect is driven primarily by a 0.9 pp reduction in remaining in the Texas workforce. There are no significant effects on employment in private schools or overall teaching employment.

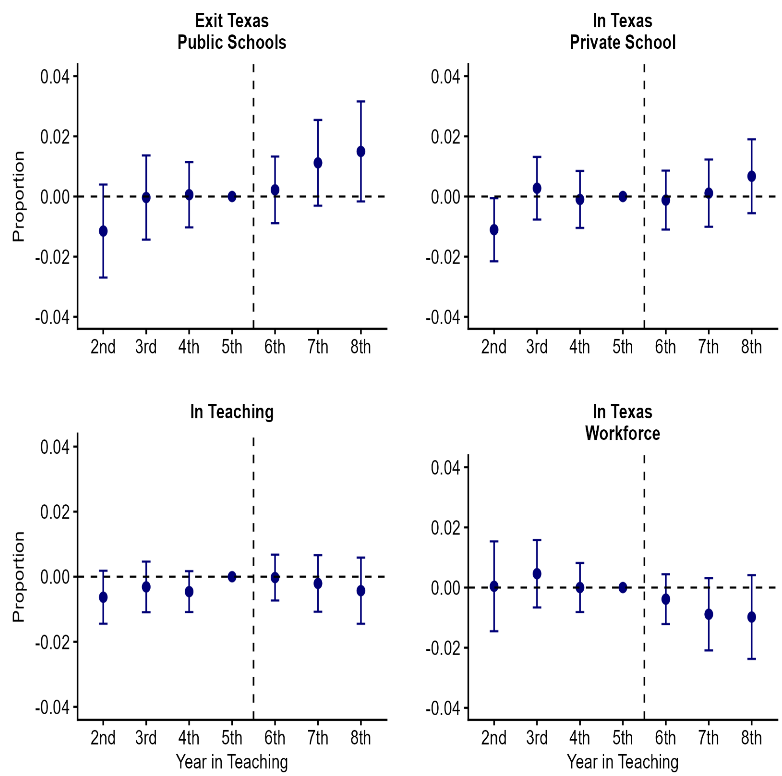
Panel A: Employment Outcomes (All)				
Outcome	**Exit TXPS**	**In Texas Private Schools**	**In Texas Workforce**	**In Teaching**
	0.012* (0.0063)	0.003 (0.0040)	-0.009* (0.0053)	0.003 (0.0040)
N	609,077	609,077	609,077	542,218
Panel B: School Outcomes (Teachers in TXPS)				
Outcome	**School Retention**	**District Retention**	**Eligible School**	**% Economically Disadvantaged**
	-0.012** (0.0057)	-0.005 (0.0045)	0.011 (0.0080)	-0.042 (0.3242)
N	483,268	483,268	483,268	483,268

Notes: Panel A reports effects on employment outcomes for all teachers in the analytic sample. Panel B conditions on remaining in TXPS and reports effects on school- and district-level retention and school characteristics. The "In Teaching" outcome is constructed only for individuals with employment reported in Texas. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table 3. Employment and School Characteristics

Figure 2 presents event study estimates for these outcomes. For exiting TXPS, pre-trends are flat, indicating similar exit trajectories over the first five years. Differences emerge after year six, with loan-eligible teachers who began in TLFP-eligible schools increasingly more likely to leave TXPS in years seven and eight. A similar dynamic appears for remaining in the Texas workforce, with declines beginning in the same period. The one-year lag may reflect the timing of forgiveness processing or the time required to secure alternative employment after reaching eligibility. Estimates for private school and overall teaching employment remain flat throughout.

Importantly, employment effects emerge only after teachers reach the point of potential for-



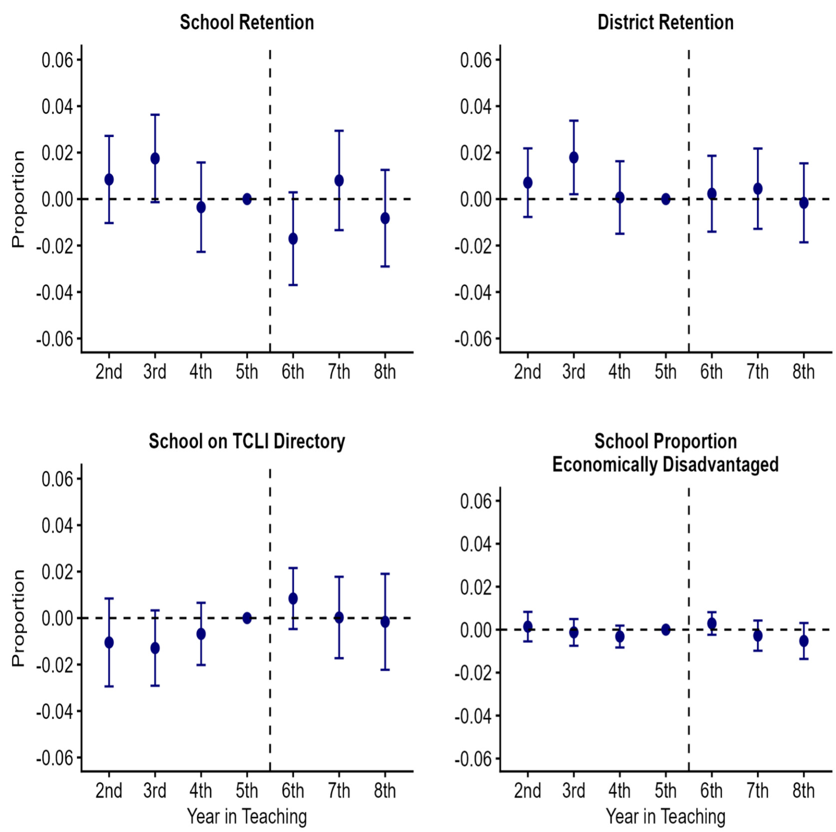
Notes: This figure contains event study estimates for teacher employment outcomes. Estimates are relative to the final year of the five-year service window (year 5).

Figure 2. Event Studies: Teacher Employment Outcomes

giveness. Teachers remain in TXPS through the service window but exit at higher rates once they become eligible for forgiveness. Because the data capture employment only within Texas, some teachers may continue teaching elsewhere. As a result, the observed declines reflect departures from TXPS rather than definitive exits from the teaching profession, in line with the null effects on remaining in teaching within Texas.

Panel B of Table 3 reports estimates for teachers who remain in TXPS, examining employment decisions within the public school system. School retention declines by 1.2 pp beginning in year six, meaning that teachers are less likely to remain in the same school after the five-year service window. As shown in Figure 3, school retention estimates fluctuate in the pre-period but do not show a clear downward trend before year five. Estimates then drop in year six, though the post-period pattern is not monotonic. However, this mobility occurs primarily within districts and is not accompanied by systematic changes in school characteristics, such as eligibility status or the share of economically disadvantaged students. Other outcomes show no statistically significant changes, indicating limited adjustments along additional within-system margins.

Overall, these results show that employment adjustments emerge after the five-year service window along two distinct margins: increased exits from TXPS and reduced school retention among those who remain. The fact that these shifts occur at the point of potential forgiveness eligibility,



Notes: This figure contains event study estimates for school/district retention and school characteristics. Estimates are relative to the final year of the five-year service window (year 5).

Figure 3. Event Study: Retention Outcomes and School Characteristics

rather than gradually over time, suggests that the TLFP affects the timing of employment decisions rather than strengthening long-run retention within TXPS. The next section examines whether adjustments extend to teachers' earnings and other employment choices.

5.2.2 Earnings and Other Employment

While much of the research on teacher loan forgiveness focuses on retention and attrition, loan forgiveness may also shape labor supply decisions among teachers who remain in public schools. By reducing student loan balances, forgiveness relaxes financial constraints and could lessen the need for additional work. At the same time, because the TLFP provides a one-time reduction that often covers only a modest share of remaining debt, these constraints are unlikely to disappear entirely. Examining earnings and employment arrangements for teachers who remain in TXPS therefore provides insight into whether the TLFP affects labor supply and employment decisions within the public school system.

To study these within-profession adjustments, I examine earnings and employment outcomes for teachers who remain in TXPS over their first eight years. These include the log of total annual earnings across all reported employment, as well as indicators for earnings from outside the public school system, holding a non-teaching position within TXPS alongside a teaching assignment, and receipt of supplemental pay. Together, these measures capture multiple margins along which teachers may adjust their labor supply while remaining in public school employment.

Outcome	**Estimate**
Log of Earnings	-0.003
	(0.0020)
Any Outside Earnings	-0.001
	(0.0056)
Non-Teaching TXPS Positions	0.001
	(0.0016)
Any Supplemental Pay	0.009
	(0.0095)
N	398,128

Notes: This table reports effects on employment outcomes for teachers who remain in TXPS for their first eight years. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table 4. Earnings and Employment Outcomes for Teachers Remaining in TXPS

Table 4 reports results for these outcomes. Overall, I find no evidence that reaching the five-year service threshold is associated with changes in earnings or employment patterns among teachers who remain in TXPS. The event-study estimates in Figure A.3 similarly show no clear pattern. In contrast to the increases in exit and declines in school retention documented earlier, potential forgiveness does not appear to alter labor supply behavior among those who stay in the public school system.

To examine whether financial motives are reflected in teachers' exit decisions, I next analyze post-exit labor market outcomes for those who leave TXPS. Changes in earnings and subsequent employment sectors illuminate whether departures are associated with movement to higher-paying opportunities or reflect transitions unrelated to earnings gains. Comparing teachers who leave before five years with those who exit after remaining more than five years provides a way to assess whether post-exit patterns differ with the timing of departure.

I implement this by adapting the DDD framework to focus on outcomes around teacher exit. For earnings, I compare changes in average annual earnings between the two years prior to exit and the two years following exit for teachers whose departure occurs before versus after five years. This approach evaluates whether exit timing relative to the service window is associated with different post-exit earnings profiles. Equation (4) outlines the specification used for this analysis:

$$\begin{aligned}\Delta Y_{is} = \alpha + TCLI_{is,t=1} + Loans_i + ExitAfter5_i + (TCLI_{is,t=1} \times Loans_i) +\\ (TCLI_{is,t=1} \times ExitAfter5_i) + (Loans_i \times ExitAfter5_i) +\\ \beta(TCLI_{is,t=1} \times Loans_i \times ExitAfter5_i) + X_{is,t=1} + ExitYear_i + \epsilon_{is}\end{aligned} \quad (4)$$

where ΔY_{is} denotes the change in average annual earnings between the two years prior to exit and the two years following exit, and $ExitAfter5_i$ is an indicator equal to one if teacher i exits TXPS after their fifth year. Using two-year averages balances concerns about short-term fluctuations against sample coverage. Longer windows would exclude teachers who exit early or near the end of the observation period, while a one-year window is more sensitive to transitory earnings variation. I report results using the one-year window as a robustness check.

As this analysis centers on exit timing, I replace the calendar-year and cohort fixed effects with exit-year fixed effects ($ExitYear_i$), which account for broader labor market conditions at the time of departure. This specification also replaces the time-varying $After5_{it}$ indicator with $ExitAfter5_i$, reflecting the focus on whether teachers exit before or after five years. The coefficient of interest, β, captures differences in post-exit earnings associated with exit timing relative to the service window.

When examining employment sector outcomes, I use the same framework but replace the earnings-change outcome with indicators for post-exit employment sector. I define sectors using NAICS two-digit industry codes, grouping post-exit employment into four mutually exclusive categories: education, public service (excluding education), private sector (excluding education), and exit from the Texas workforce. Education includes employment in educational services (NAICS 61); public service includes health care and social assistance, other service organizations, and public administration (NAICS 62, 81, and 92); private sector employment includes all remaining NAICS two-digit industries; and exit from the Texas workforce is defined as having no reported earnings in Texas. For teachers with earnings in multiple sectors following exit, I classify their primary sector in each post-exit year (year one and year two separately) as the sector in which they report the highest total earnings in that year.

Table 5 reports estimates for post-exit employment sector and earnings outcomes.[6] Teachers who exit after the five-year service threshold are modestly less likely to remain in education in the two-year window (5.2 pp), but there is no corresponding increase in transitions into public service or private sector employment, nor evidence of increased exits from the Texas workforce. Earnings estimates similarly show small and statistically insignificant differences across both the two-year and one-year windows.[7] Overall, these results provide little evidence that exits occurring after the five-year service window reflect systematic movement toward higher-paying or broad shifts across sectors.

5.3 Heterogeneity

While the results presented above show how potential eligibility shapes employment decisions after five years, these effects may conceal important heterogeneity across teachers. Eligibility for loan forgiveness may interact with teachers' existing financial constraints and career opportunities. For instance, differences in loan balances, forgiveness amounts, and outside labor-market options may

[6] See Appendix Figures A.4-A.6 for event studies for these outcomes.

[7] Sample sizes differ across sector and earnings outcomes because earnings changes are only observed for teachers with reported earnings in both the pre- and post-exit windows.

Outcome	2 Years	1 Year
Primary Sector: Education	-0.052*	-0.004
	(0.0288)	(0.0234)
Primary Sector: Public	0.002	0.000
	(0.0114)	(0.0080)
Primary Sector: Private	0.000	-0.011
	(0.0176)	(0.0141)
Out of Texas Workforce	0.049	0.016
	(0.0298)	(0.0249)
Teachers	26,562	36,999
Change in Earnings	-1167	-1455
	(2459)	(1433)
N	15,139	33,295

Notes: This table reports effects on primary sector of employment and change in earnings for teachers who ever leave TXPS. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table 5. Sector and Earnings Outcomes Leavers

shape how teachers respond. Initial school environments and individual characteristics may further mediate these responses.

To examine this heterogeneity, I extend the baseline DDD specification shown in Equation 2 to allow the post-five-year effect of initial placement in a TLFP-eligible school to vary across groups. Equation 5 shows the specification used for this analysis:

$$\begin{aligned} Y_{ist} = \alpha + DDD_{ist} + Group_i + (DDD_{ist} \times Group_i) + \\ X_{is,t=1} + \delta_{\text{Cohort}(i)} + \lambda_t + \epsilon_{ist} \end{aligned} \tag{5}$$

where DDD_{ist} contains the baseline DDD terms from Equation 2, the coefficient on the triple-interaction ($TCLI_{is,t=1} \times Loans_i \times After5_{it}$) captures the post-five-year effect for the reference group, and its interaction with $Group_i$ captures how this effect differs across groups. I alter Equation 4 in a similar manner to explore the earnings and employment sector of those who leave the TXPS system.

I define the group indicator $Group_i$ by assigning teachers to mutually exclusive categories within each of the following teacher- and school-level dimensions:

1. *Student loan balance*: Less than \$10,000 (reference), \$10,001–\$20,000, \$20,001–\$35,000, \$35,001–\$55,000, \$55,001–\$80,000, and \$80,001+, corresponding to the 10th, 25th, 50th, 75th, and 90th percentiles of the distribution.
2. *Forgiveness amount eligibility*: Eligibility for \$5,000 in loan forgiveness (reference) or \$17,500, with the \$17,500 category further split into special education and secondary mathematics/science teachers.

3. *Certification route*: Traditional certification (reference), post-baccalaureate certification, or alternative certification
4. *Race/ethnicity*: White (reference), Black, Hispanic/Latino, Asian, and other race/ethnicity categories.
5. *Initial school economic disadvantage*: Less than 50% economically disadvantaged (reference), [50%, 70%), [70%, 90%) and 90%+.

5.3.1 Employment Outcomes

I begin by examining how employment outcomes vary with teachers' student loan balances. Panel A of Table A.1 shows that exit from TXPS generally increases with loan balance, with the largest and only statistically significant effect for teachers holding between $55,000 and $80,000 in eligible loans (4.0 pp relative to those with less than $10,000). In contrast, there is little evidence of differential movement into private schools or continued teaching across balance categories. Instead, teachers with larger loan balances—particularly those above $35,000—are significantly less likely to remain in the Texas workforce. This pattern suggests that reaching the five-year service threshold loosens the tie to public schools, and that teachers with substantial debt may use that flexibility to make broader labor-market or geographic transitions rather than remaining in-state.

Panel B examines heterogeneity by the economic disadvantage of teachers' initial schools. Teachers whose initial placements fell below 50% economically disadvantaged enrollment are 1.5 pp more likely to leave TXPS after five years. Exit rates for teachers who began in schools with higher levels of economic disadvantage are not different from those in schools below 50%. Across the remaining outcomes, there is no evidence of meaningful differences in private school employment or continued teaching. Similarly, teachers from schools below 50% economically disadvantaged enrollment are slightly less likely to remain in the Texas workforce, while estimates for higher-disadvantage schools are statistically indistinguishable from zero.

Panels C and D examine heterogeneity by forgiveness amount, subject area, and certification route. Across these dimensions, there is little evidence that employment responses differ meaningfully from the baseline effects. Teachers eligible for larger $17,500 forgiveness amounts do not exhibit statistically different exit or employment patterns relative to those eligible for $5,000. Similarly, employment outcomes do not vary substantially by certification pathway.

In contrast, Panel E reveals meaningful heterogeneity by race/ethnicity. Relative to White teachers, Black teachers are 13 pp less likely to exit TXPS after five years and are also less likely to transition into private schools (3.8 pp). At the same time, they are 7.6 pp more likely to remain in the Texas workforce. Asian teachers exhibit a similar pattern in exit behavior, with a 7.8 pp lower likelihood of leaving TXPS, and are also more likely to remain in the Texas workforce (8.6 pp). Estimates for Hispanic/Latino teachers and teachers from other races/ethnicities are not different from those of White teachers across these outcomes.

5.3.2 Earnings and Other Employment Outcomes (Stayers)

I next examine heterogeneity in earnings and within-profession employment adjustments among teachers who remain in TXPS, with results reported in Table A.2. Panels A and B show little variation by loan balance or by the economic disadvantage of teachers' initial schools. Across these dimensions, estimates for total earnings, outside earnings, and non-teaching roles are largely similar. The only difference appears in supplemental pay, where teachers who began in schools with 50–90% economically disadvantaged enrollment are 1.8-1.9 pp less likely to receive supplemental pay after five years relative to those who started in schools below 50%.

Differences by forgiveness amount and subject area are modest (Panel C). Secondary science and mathematics teachers experience a 1.1% increase in total earnings and are 5.3 pp more likely to report outside earnings relative to the $5,000 reference group, while special education teachers show no comparable adjustments. Variation by certification route is similarly limited (Panel D), though teachers certified through a post-baccalaureate program exhibit a small increase in total earnings (1.4%), whereas alternatively certified teachers are less likely to receive supplemental pay.

A more distinct pattern emerges by race and ethnicity. In Panel E, total earnings do not differ significantly across groups after five years, but Black teachers are 10.4 pp less likely to hold outside earnings and 11.4 pp less likely to receive supplemental pay relative to White teachers. These results are consistent with a reduction in multiple job holding among Black teachers after reaching the five-year service threshold, even though overall earnings do not decline.

5.3.3 Earnings and Other Employment Outcomes (Leavers)

Tables A.3 and A.4 examine heterogeneity in post-exit sector and earnings outcomes among teachers who leave TXPS. Overall, there is limited evidence of systematic differences in post-exit employment sector across loan balance, initial school context, forgiveness amount, subject area, or certification route. While a few individual estimates are statistically significant, effects are generally small and rarely match across the one- and two-year windows. One exception appears among special education teachers, who are 20-25 pp less likely to be out of the Texas workforce following exit. Beyond this case, sectoral transitions after leaving TXPS do not vary meaningfully across groups.

Earnings outcomes tell a similar story. Across all panels and both time windows, none of the estimated differences in post-exit earnings are statistically significant, providing no evidence of systematic variation across groups.

6 Conclusion

This paper examines how teachers respond to potential eligibility for student loan forgiveness under the TLFP, with particular attention to employment decisions after the program's five-year service requirement. Using detailed administrative data from Texas and a DDD framework that exploits variation in teacher loan eligibility and initial school eligibility, I find no evidence that the TLFP increases retention during the five-year service window or produces sustained improvements in long-term attachment to TXPS. Instead, potential eligibility shifts the timing of exit, with teachers more likely to leave TXPS after satisfying the five-year requirement, particularly among those with large student loan balances. However, because the data capture employment only within Texas, these estimates reflect exits from TXPS rather than definitive exits from the teaching profession, as some teachers may continue teaching in other states that are not observed in these data.

For teachers who remain in TXPS, I find evidence of reduced school-level retention after the five-year service window. At the same time, this movement does not correspond to systematic changes in the types of schools in which teachers work, nor is it accompanied by differences in earnings, supplemental pay, or employment outside the classroom. This pattern indicates that post-service adjustments within TXPS are limited to school switching rather than broader changes in placement or compensation.

Heterogeneity analyses reveal that responses to loan forgiveness eligibility vary across certain teacher and institutional characteristics. Teachers with mid-to-high student loan balances are more likely to leave TXPS after five years, consistent with debt relief playing a larger role in shaping employment decisions for those with higher balances. There are also differences by race/ethnicity, with Black teachers substantially less likely to exit after five years and, among those who remain, less likely to engage in outside earnings or supplemental work after reaching the service threshold.

These findings have several implications for the design of teacher retention policies. First, one-time, service-contingent loan forgiveness appears insufficient to promote sustained retention in public schools. Even relatively large forgiveness amounts may be small relative to teachers' remaining debt burdens and alternative career opportunities, limiting their ability to support long-term retention beyond the service period. Second, policies that rely on discrete eligibility thresholds may be associated with elevated exits once service obligations are met. Retention strategies that provide more continuous financial support—such as incremental loan forgiveness or compensation paths that evolve with experience—may therefore be better suited to stabilizing the teacher workforce over time.

Several limitations of this study point to directions for future research. Because the administrative data used in this study do not identify which teachers apply for or receive loan forgiveness, the estimates reflect the effects of potential eligibility rather than program take-up. Linking forgiveness receipt directly to employment outcomes would allow for a more direct assessment of the program's effectiveness in shaping teacher retention and career trajectories. This is particularly important given the scale of federal investment in teacher loan forgiveness programs, which involves billions of dollars in loan forgiveness. Future work that links administrative records across states or data systems would also help distinguish exits from specific public school systems from continued employment in public schools elsewhere. Better data on both forgiveness receipt and teachers' employment across public school systems would substantially improve policymakers' ability to evaluate whether these expenditures meaningfully influence teacher behavior.

Future work should extend this analysis along several substantive dimensions. While this paper focuses on employment and earnings outcomes, additional research should examine impacts on teacher quality, including whether exits after the service period disproportionately involve more or less effective teachers. More broadly, comparative evaluations of alternative program designs would be especially valuable. For example, programs that provide incremental loan forgiveness or repayment assistance over longer horizons, as well as policies such as the TEACH Grant that condition aid on continued service rather than forgiving existing debt, may generate different retention responses. Extending similar analyses to other states and policy contexts would help clarify which features of financial incentive programs are most effective in supporting long-term teacher retention.

Overall, the evidence suggests that while loan forgiveness programs like the TLFP may influence the timing of teacher exits, they are unlikely to resolve persistent teacher shortages on their own. Addressing long-term retention will likely require policies that go beyond one-time financial relief and instead confront the broader financial and career constraints facing teachers throughout their careers.

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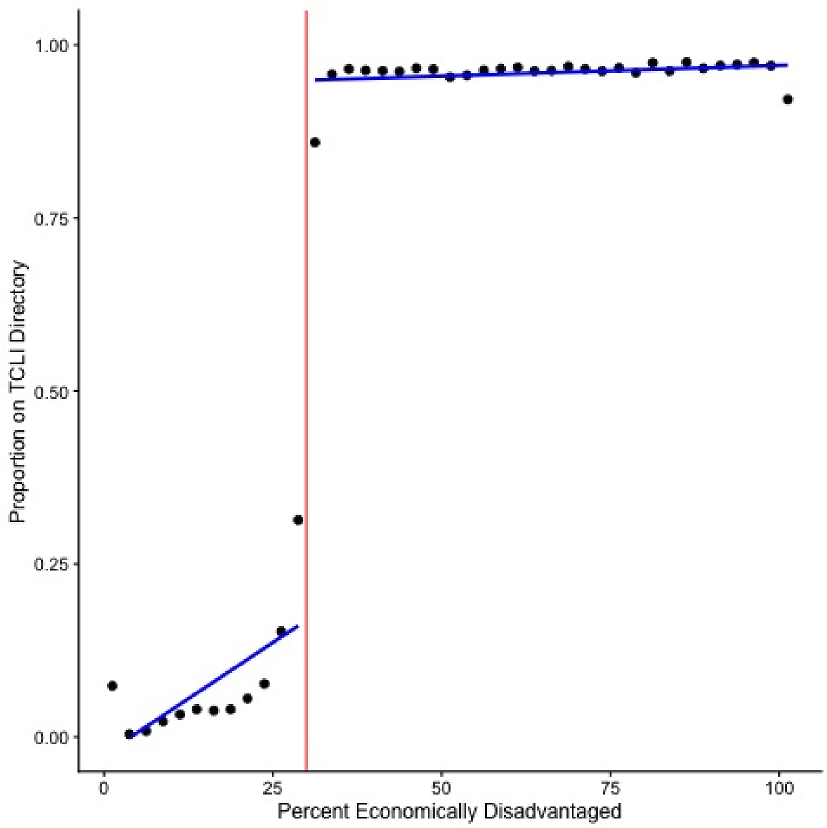
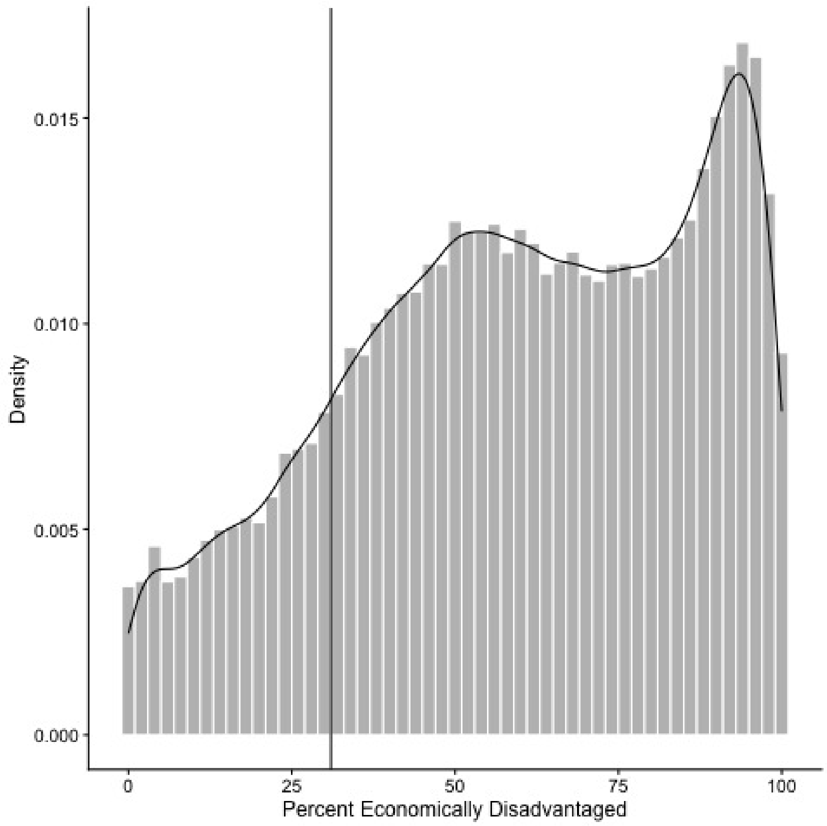
A Appendix



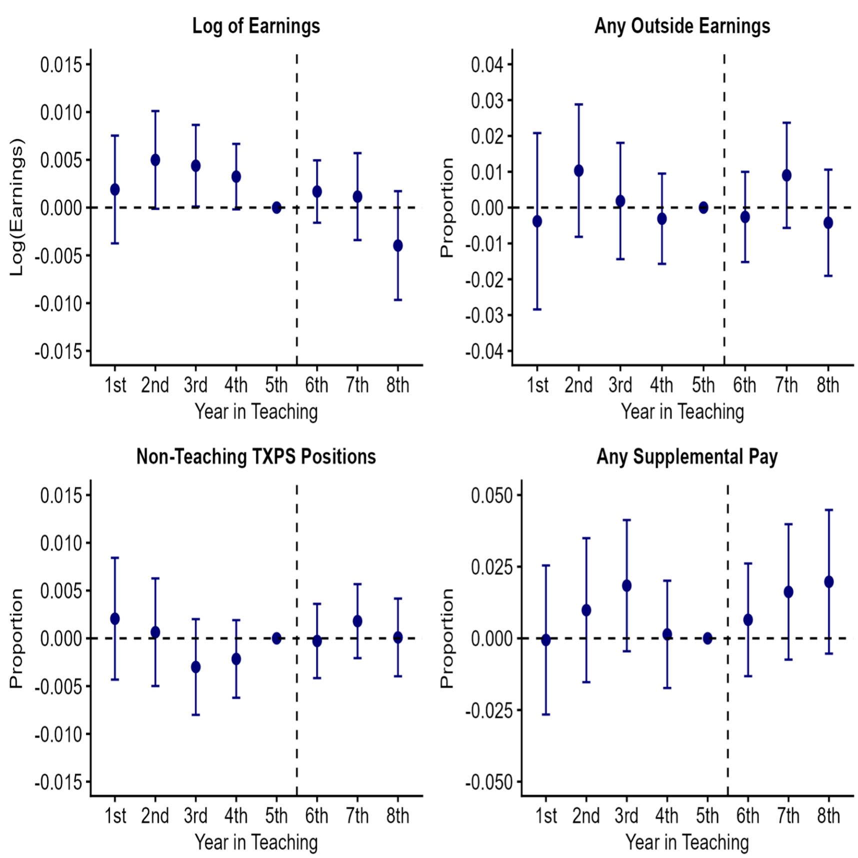
Notes: This figure shows the percent of schools in each bin of width 2.5 that are on the TCLI Directory that qualifies a teacher's year in teaching toward the TLFP service requirement.

Figure A.1. First Stage



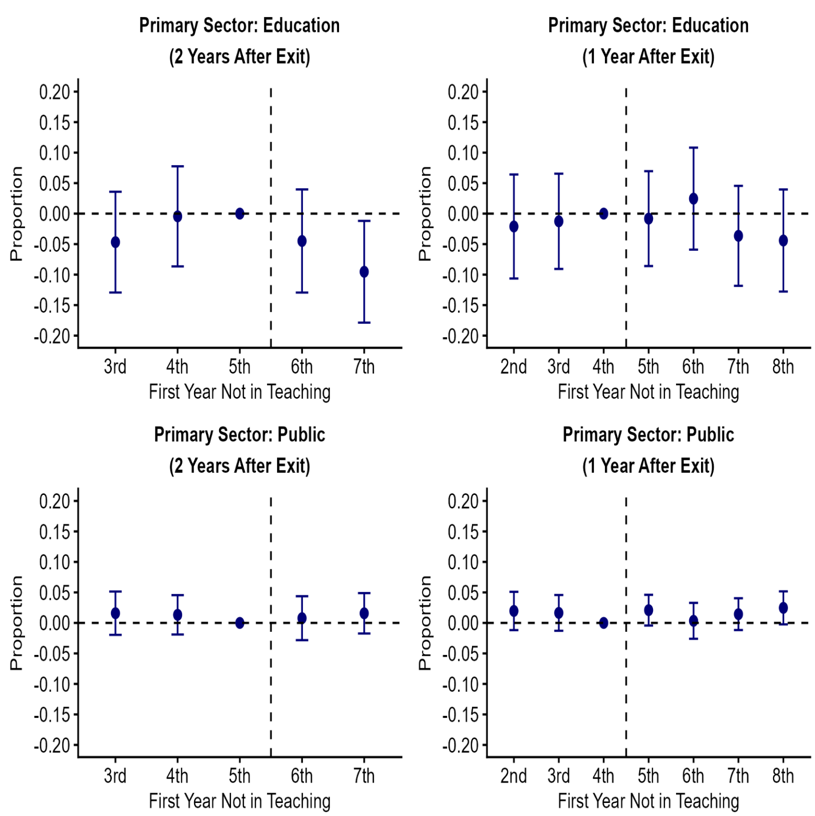
Notes: This figure shows the distribution of schools by the percent of their total enrollment designated as economically disadvantaged.

Figure A.2. Distribution of School Percent Economically Disadvantaged



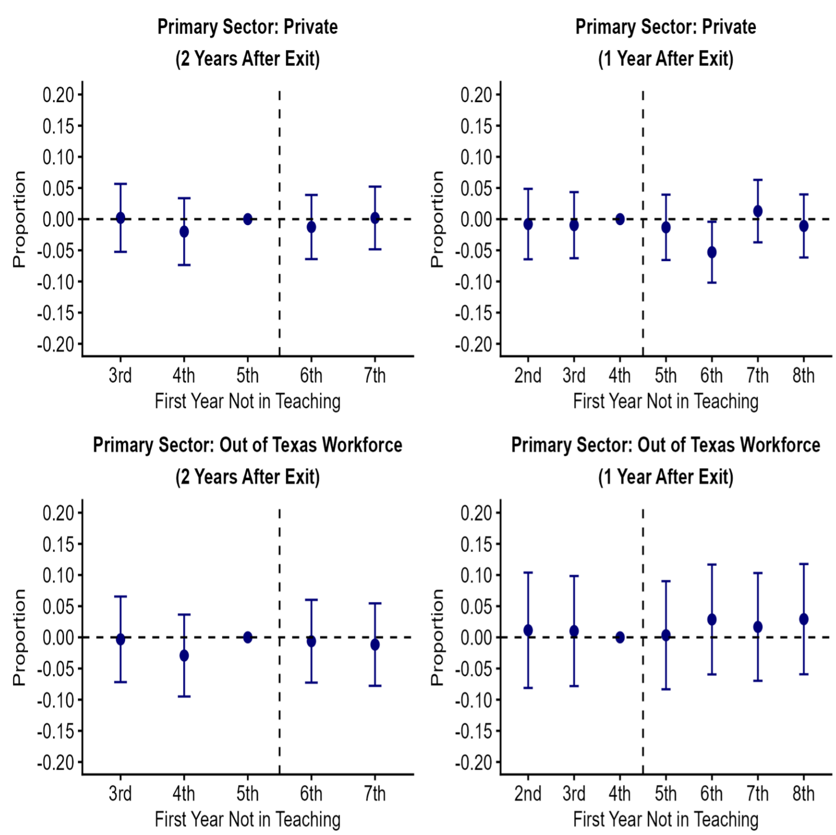
Notes: This figure contains event study estimates for earnings and other employment outcomes for teachers remaining in TXPS. Estimates are relative to the final year of the five-year service window (year 5).

Figure A.3. Event Study: Earnings and Employment Outcomes for Teachers Remaining in TXPS



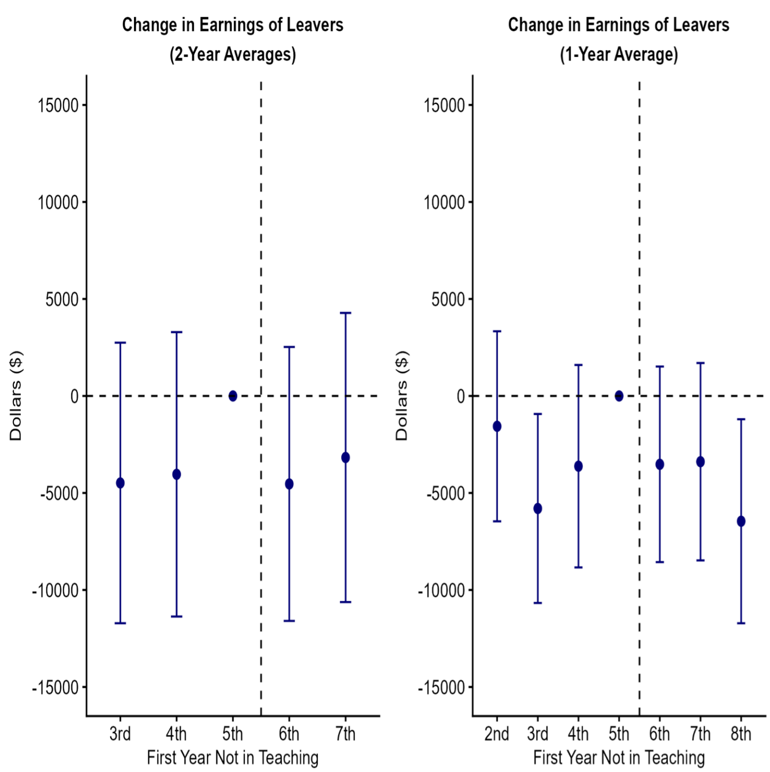
Notes: This figure contains event study estimates for primary sector of employment for individuals who leave TXPS. Estimates are relative to the final year of the five-year service window (year 5).

Figure A.4. Event Study: Primary Sector of Employment for Leavers



Notes: This figure contains event study estimates for primary sector of employment for individuals who leave TXPS. Estimates are relative to the final year of the five-year service window (year 5).

Figure A.5. Event Study: Primary Sector of Employment for Leavers



Notes: This figure contains event study estimates for earnings for individuals who leave TXPS. Estimates are relative to the final year of the five-year service window (year 5).

Figure A.6. Event Study: Change in Earnings for Leavers

Panel A: Eligible Loans	(1) Exit TXPS	(2) In Texas Private Schools	(3) In Teaching	(4) In Texas Workforce
Reference: <$10,000	-0.006	0.004	-0.005	0.010
	(0.0136)	(0.0087)	(0.0085)	(0.0111)
$10,001-$20,000	-0.012	-0.001	0.024**	-0.006
	(0.0167)	(0.0104)	(0.0106)	(0.0135)
$20,001-$35,000	0.017	0.000	0.002	-0.014
	(0.0149)	(0.0096)	(0.0089)	(0.0121)
$35,001-$55,000	0.023	-0.001	0.005	-0.025**
	(0.0149)	(0.0096)	(0.0087)	(0.0120)
$55,001-$80,000	0.040***	0.000	0.005	-0.042***
	(0.0160)	(0.0109)	(0.0097)	(0.0124)
$80,001+	0.024	-0.005	0.012	-0.038***
	(0.0178)	(0.0128)	(0.0114)	(0.0130)
Panel B: Initial School % Economically Disadvantaged				
Reference: <50%	0.015*	0.005	0.005	-0.012*
	(0.0079)	(0.0051)	(0.0052)	(0.0067)
50% to <70%	-0.008	-0.005	-0.007	0.007
	(0.0072)	(0.0049)	(0.0047)	(0.0059)
70% to <90%	0.003	0.004	-0.006	0.001
	(0.0073)	(0.0049)	(0.0045)	(0.0058)
90%+	0.000	-0.005	-0.006	-0.003
	(0.0087)	(0.0062)	(0.0049)	(0.0066)
Panel C: Forgiveness Amount				
Reference: $5,000	0.013*	0.00	0.003	-0.010*
	(0.0068)	(0.0043)	(0.0041)	(0.0057)
$17,500, Special Education	-0.007	-0.023	0.005	-0.012
	(0.0342)	(0.0239)	(0.0215)	(0.0287)
$17,500, Secondary Science/Mathematics	0.005	0.004	-0.011	0.007
	(0.0219)	(0.0150)	(0.0155)	(0.0174)
Panel D: Certification Route				
Reference: Traditional	0.017**	0.007	-0.001	-0.009
	(0.0067)	(0.0042)	(0.0042)	(0.0057)
Post-Bac	-0.033	-0.018	0.019	-0.013
	(0.0247)	(0.0151)	(0.0190)	(0.0224)
Alternative	-0.035	-0.010	0.032*	0.007
	(0.0282)	(0.0189)	(0.0180)	(0.0228)
Panel E: Race/Ethnicity				
Reference: White	0.011	0.001	0.002	-0.012*
	(0.0071)	(0.0044)	(0.0046)	(0.0061)
Hispanic/Latino	0.001	0.001	-0.009	0.009
	(0.0213)	(0.0134)	(0.0113)	(0.0177)
Black	-0.130***	-0.038*	0.013	0.076***
	(0.0311)	(0.0226)	(0.0184)	(0.0238)
Asian	-0.078*	0.029	0.020	0.086**
	(0.0424)	(0.0293)	(0.0352)	(0.0395)
Other	-0.028	0.002	0.016	0.026
	(0.0542)	(0.0336)	(0.0368)	(0.0453)
N	609,077	609,077	542,218	609,077

Notes: This table reports effects on employment outcomes for all teachers in the analytic sample. Panel A reports estimates by loan balance, Panel B by initial school % economically disadvantaged, Panel C by forgiveness amount and subject, Panel D by certification route, and Panel E by race/ethnicity. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table A.1. Employment Heterogeneity

Panel A: Eligible Loans	**(1)** **Log of Earnings**	**(2)** **Any Outside Earnings**	**(3)** **Non-Teaching TXPS Position**	**(4)** **Any Supplemental Pay**
Reference: < $10,000	-0.001	0.006	0.001	-0.001
	(0.0039)	(0.0112)	(0.0041)	(0.0190)
$10,001-$20,000	0.000	-0.018	0.000	-0.004
	(0.0049)	(0.0136)	(0.0052)	(0.0231)
$20,001-$35,000	0.000	-0.003	0.001	-0.001
	(0.0045)	(0.0125)	(0.0042)	(0.0207)
$35,001-$55,000	-0.004	-0.017	-0.002	0.014
	(0.0042)	(0.0125)	(0.0045)	(0.0198)
$55,001-$80,000	-0.004	0.002	-0.001	0.021
	(0.0048)	(0.0140)	(0.0049)	(0.0224)
$80,001+	0.001	0.002	0.000	0.034
	(0.0051)	(0.0161)	(0.0050)	(0.0246)
Panel B: Initial School % Economically Disadvantaged				
Reference: <50%	0.000	-0.001	0.001	0.022*
	(0.0026)	(0.0070)	(0.0020)	(0.0119)
50% to <70%	-0.001	0.001	0.001	-0.019*
	(0.0026)	(0.0064)	(0.0016)	(0.0106)
70% to <90%	-0.005*	-0.001	-0.001	-0.018*
	(0.0024)	(0.0061)	(0.0016)	(0.0104)
90%+	0.001	-0.001	0.001	-0.009
	(0.0028)	(0.0069)	(0.0016)	(0.0119)
Panel C: Forgiveness Amount				
Reference: $5,000	-0.004**	-0.007	0.001	0.007
	(0.0021)	(0.0060)	(0.0017)	(0.0100)
$17,500, Special Education	0.003	0.041	-0.017	0.038
	(0.0108)	(0.0346)	(0.0110)	(0.0541)
$17,5000, Secondary Science/Mathematics	0.011*	0.053***	0.000	-0.008
	(0.0070)	(0.0204)	(0.0062)	(0.0335)
Panel D: Certification Route				
Reference: Traditional	-0.003	-0.003	0.001	0.012
	(0.0021)	(0.0058)	(0.0017)	(0.0101)
Post-Bac	0.014*	0.014	-0.001	0.012
	(0.0073)	(0.0237)	(0.0076)	(0.0373)
Alternative	-0.012	0.011	-0.002	-0.096**
	(0.0089)	(0.0233)	(0.0070)	(0.0425)
Panel E: Race/Ethnicity				
Reference: White	-0.001	-0.001	0.002	0.012
	(0.0023)	(0.0062)	(0.0019)	(0.0107)
Hispanic/Latino	-0.007	0.006	-0.005	-0.016
	(0.0071)	(0.0206)	(0.0042)	(0.0334)
Black	-0.007	-0.104**	-0.005	-0.114*
	(0.0136)	(0.0518)	(0.0076)	(0.0685)
Asian	-0.004	-0.021	0.000	0.042
	(0.0141)	(0.0457)	(0.0066)	(0.0600)
Other	0.005	0.002	0.002	-0.062
	(0.0133)	(0.0529)	(0.0062)	(0.0855)
N	398,128	398,128	398,128	398,128

Notes: This table reports effects on employment outcomes for teachers who remain in TXPS for their first eight years. Panel A reports estimates by loan balance, Panel B by initial school % economically disadvantaged, Panel C by forgiveness amount and subject, Panel D by certification route, and Panel E by race/ethnicity. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table A.2. Earnings Heterogeneity: Stayers

	2 Years				1 Year			
Primary Employment Sector	**Education**	**Public**	**Private**	**Out of TXWF**	**Education**	**Public**	**Private**	**Out of TXWF**
Panel A: Eligible Loans								
Reference: < $10,000	-0.139**	0.007	0.013	0.005	-0.013	-0.001	-0.049	-0.011
	(0.0616)	(0.0258)	(0.0383)	(0.0513)	(0.0494)	(0.0181)	(0.0306)	(0.0432)
$10,001-$20,000	0.105	-0.035	-0.024	0.005	-0.004	-0.020	0.024	0.026
	(0.0739)	(0.0353)	(0.0473)	(0.0628)	(0.0597)	(0.0229)	(0.0375)	(0.0525)
$20,001-$35,000	0.091	-0.003	0.009	-0.009	0.006	-0.003	0.054	0.003
	(0.0671)	(0.0279)	(0.0409)	(0.0560)	(0.0550)	(0.0199)	(0.0344)	(0.0464)
$35,001-$55,000	0.129*	0.008	-0.027	-0.005	0.049	0.008	0.023	0.007
	(0.0686)	(0.0292)	(0.0419)	(0.0554)	(0.0552)	(0.0201)	(0.0336)	(0.0465)
$55,001-$80,000	0.102	-0.010	-0.030	0.021	0.020	0.007	0.063	0.018
	(0.0792)	(0.0312)	(0.0469)	(0.0630)	(0.0640)	(0.0215)	(0.0387)	(0.0532)
$80,001+	0.023	0.008	-0.003	-0.015	-0.086	0.009	0.060	0.002
	(0.0825)	(0.0356)	(0.0491)	(0.0645)	(0.0685)	(0.0241)	(0.0380)	(0.0551)
Panel B: Initial School % Economically Disadvantaged								
Reference: <50%	-0.039	0.002	-0.014	0.000	0.006	-0.001	-0.006	0.000
	(0.0363)	(0.0146)	(0.0238)	(0.0281)	(0.0297)	(0.0103)	(0.0194)	(0.0323)
50% to <70%	-0.025	0.002	0.011	0.007	-0.025	0.001	-0.018	0.045
	(0.0351)	(0.0137)	(0.0226)	(0.0269)	(0.0284)	(0.0108)	(0.0188)	(0.0310)
70% to <90%	0.013	0.007	0.015	-0.027	-0.004	0.008	-0.009	0.006
	(0.0349)	(0.0145)	(0.0225)	(0.0266)	(0.0284)	(0.0106)	(0.0183)	(0.0311)
90%+	-0.063	-0.019	0.039	0.042	-0.008	-0.005	0.022	-0.003
	(0.0438)	(0.0177)	(0.0248)	(0.0334)	(0.0359)	(0.0128)	(0.0201)	(0.0380)
Panel C: Forgiveness Amount								
Reference: $5,000	-0.058*	0.001	-0.002	0.012	-0.006	0.005	-0.014	0.007
	(0.0302)	(0.0121)	(0.0187)	(0.0238)	(0.0248)	(0.0085)	(0.0150)	(0.0200)
$17,500, Special Education	0.129	0.047	0.109	-0.247***	0.014	-0.059	0.094	-0.205***
	(0.1446)	(0.0735)	(0.0716)	(0.0871)	(0.1242)	(0.0576)	(0.0744)	(0.0796)
$17,5000, Secondary Science/Mathematics	0.024	-0.015	-0.033	-0.004	-0.008	-0.042	0.008	-0.023
	(0.1098)	(0.0357)	(0.0697)	(0.0836)	(0.0859)	(0.0259)	(0.0565)	(0.0692)
Panel D: Certification Route								
Reference: Traditional	-0.050	0.004	-0.007	0.010	-0.043*	0.001	-0.016	0.008
	(0.0307)	(0.0122)	(0.0186)	(0.0240)	(0.0258)	(0.0086)	(0.0148)	(0.0265)
Post-Bac	-0.086	0.002	0.016	-0.135	-0.020	-0.002	0.010	0.166
	(0.1215)	(0.0436)	(0.0754)	(0.0907)	(0.1032)	(0.0320)	(0.0635)	(0.1101)
Alternative	0.056	-0.018	0.116	0.002	0.089	-0.003	0.093	-0.044
	(0.1202)	(0.0497)	(0.0801)	(0.0976)	(0.0984)	(0.0380)	(0.0613)	(0.1024)
Panel E: Race/Ethnicity								
Reference: White	-0.030	0.000	-0.001	-0.013	0.010	-0.003	-0.013	0.006
	(0.0315)	(0.0123)	(0.0197)	(0.0250)	(0.0253)	(0.0089)	(0.0160)	(0.0274)
Hispanic/Latino	-0.206*	0.031	0.114*	0.023	-0.123	0.008	0.072	0.062
	(0.1075)	(0.0382)	(0.0647)	(0.0763)	(0.0878)	(0.0222)	(0.0538)	(0.0932)
Black	0.577*	0.026	-0.136**	0.091	0.151	0.059	-0.049	-0.162
	(0.3137)	(0.0687)	(0.0672)	(0.1044)	(0.4221)	(0.0452)	(0.0634)	(0.4532)
Asian	0.117	-0.001	0.025	0.0232	-0.035	0.041	-0.029	0.018
	(0.1949)	(0.0746)	(0.1097)	(0.1586)	(0.1585)	(0.0602)	(0.0880)	(0.1668)
Other	-0.109	0.127	-0.158	-0.136	-0.302	-0.059	-0.082	0.397*
	(0.2389)	(0.1348)	(0.1295)	(0.1476)	(0.2073)	(0.0874)	(0.1021)	(0.2247)
N	26,562	26,562	26,562	26,562	36,999	36,999	36,999	36,999

Notes: This table reports effects on primary sector of employment for teachers who ever leave TXPS. Panel A reports estimates by loan balance, Panel B by initial school % economically disadvantaged, Panel C by forgiveness amount and subject, Panel D by certification route, and Panel E by race/ethnicity. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table A.3. Sector Heterogeneity: Leavers

Panel A: Eligible Loans	2-Years	1-Year
Reference: < $10,000	-3919	-2935
	(4754)	(2862)
$10,001-$20,000	9827	807
	(6117)	(3954)
$20,001-$35,000	3007	3528
	(5214)	(3194)
$35,001-$55,000	3298	1024
	(5129)	(3254)
$55,001-$80,000	727	3011
	(5443)	(3712)
$80,001+	-914	-3664
	(5795)	(4022)
Panel B: Initial School % Economically Disadvantaged		
Reference: <50%	-690	-500
	(3163)	(2056)
50% to <70%	-3688	-2732
	(2857)	(1969)
70% to <90%	1824	-1206
	(2850)	(1942)
90%+	735	1397
	(3365)	(2356)
Panel C: Forgiveness Amount		
Reference: $5,000	-909	-830
	(2717)	(1566)
$17,500, Special Education	6045	-1982
	(10721)	(7640)
$17,5000, Secondary Science/Mathematics	-6083	-7805
	(9105)	(5606)
Panel D: Certification Route		
Reference: Traditional	-1013	-1413
	(2605)	(1525)
Post-Bac	-882	-3067
	(10544)	(6353)
Alternative	-5627	497
	(11255)	(5884)
Panel E: Race/Ethnicity		
Reference: White	-1193	-1307
	(2715)	(1566)
Hispanic/Latino	1069	913
	(8819)	(5367)
Black	-4304	2741
	(8853)	(6422)
Asian	-3222	-5440
	(18973)	(11452)
Other	-2844	-1486
	(22927)	(12027)
N	15,139	33,295

Notes: This table reports effects on change in earnings for teachers who ever leave TXPS. Panel A reports estimates by loan balance, Panel B by initial school % economically disadvantaged, Panel C by forgiveness amount and subject, Panel D by certification route, and Panel E by race/ethnicity. Standard errors clustered at the initial school are in parentheses. *p<0.10; **p<0.05; ***p<0.01.

Table A.4. Earnings Heterogeneity: Leavers